



PD Tasks Team - Synthesis

Frequency Sweep Synthesis - RV12 on ASAP7

Week 04 - Synthesis at 500 / 600 / 700 MHz with RV12 Analysis

Design: RV12 - RoaLogic RISC-V CPU with AHB3-Lite interface

PDK / Library: ASAP7 7nm predictive - asap7sc7p5t_AO RVT

Tool flow: OpenROAD-flow-scripts (Docker openroad/orfs) - Yosys + ABC - OpenROAD STA

Clock targets: 500 MHz (2.000 ns) - 600 MHz (1.667 ns) - 700 MHz (1.429 ns) on port HCLK

GitHub Repository: Fardeen2228/rv12-npu-synthesis-final

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Table of Contents

1. Overview and Toolchain Note	3
2. Repository and Deliverable Check	3
3. RTL Preparation and Design Selection	4
4. Clock Constraints and Design Configuration	4
5. Issues Faced and How Each Was Fixed	5
6. Running Synthesis at All Three Frequencies	6
7. Metrics Extraction and Sources	6
8. Day 1 Metrics Table	7
9. Getting STA Numbers - make floorplan	7
10. Critical Path Analysis and Maximum Frequency	8
11. RV12 Design-Specific Challenges and Questions	9
12. Frequency Scaling Observations	10
13. Deliverables and GitHub Submission Packaging	10
14. Key Facts Learned This Week	11
15. Reflection and Recommendation / Future Scope	11

1. Overview and Toolchain Note

This report documents the Week 4 frequency-sweep synthesis of the RV12 RISC-V CPU at 500 MHz, 600 MHz, and 700 MHz on the ASAP7 7nm predictive platform. The synthesis top used in the final run was `riscv_top_ahb3lite`, with HCLK as the clock port. For each frequency, I extracted the required structural and timing metrics: cell count, chip area, flip-flop count, most common cell type, WNS, TNS, and the critical path.

The assignment was originally approached through OpenLane and Sky130 during earlier setup practice, but the final Week 4 results in this report use OpenROAD-flow-scripts with ASAP7. ORFS provides the ASAP7 platform and the same core engines needed for synthesis and timing analysis: Yosys, ABC mapping, OpenROAD database generation, and OpenSTA timing reports.

Item	Expected final requirement	Used in this report
Platform / library	ASAP7 7nm, asap7sc7p5t	ASAP7, asap7sc7p5t_AO RVT
Frequencies	500 / 600 / 700 MHz	500 / 600 / 700 MHz
Clock periods	2.000 / 1.667 / 1.429 ns	2000 / 1667 / 1429 ps in SDC
Tool flow	Synthesis + timing evidence	ORFS Docker openroad/orfs
Top / clock	Design-specific	riscv_top_ahb3lite / HCLK

2. Repository and Deliverable Check

I verified the GitHub repository after pushing the final RV12 data. The repository contains both the `npu/` and `rv12/` deliverable folders, and the README now states that both designs were synthesized using OpenROAD-flow-scripts with ASAP7 at 500, 600, and 700 MHz. The `rv12/` folder contains configs, constraints, logs, reports, results/netlist, and `RV12_frequency_analysis.md`. The `npu/` folder has the equivalent evidence plus the final NPU report folder.

Deliverable	Status	Evidence folder/file
RV12 configs	Uploaded	rv12/configs/config_500.mk, config_600.mk, config_700.mk
RV12 constraints	Uploaded	rv12/constraints/constraint_500.sdc, constraint_600.sdc, constraint_700.sdc
RV12 synthesis reports	Uploaded	rv12/reports/synthesis/<frequency>/synth_stat.txt
RV12 floorplan STA reports	Uploaded	rv12/reports/floorplan/<frequency>/2_floorplan_final.rpt
RV12 netlists	Uploaded	rv12/results/netlist/<frequency>/rv12_<frequency>.v
Summary analysis	Uploaded	rv12/RV12_frequency_analysis.md

Note: The repository still shows older synthesis_200/300/400 folders from earlier OpenLane/Sky130 practice. The final Week 4 ASAP7 deliverables are the configs, constraints, reports, logs, results/netlist, and RV12_frequency_analysis.md folders/files listed above.

3. RTL Preparation and Design Selection

The RV12 source had already been converted into a single synthesized Verilog file from the earlier OpenLane setup. I reused the corrected converted RTL, staged it inside the ORFS source tree, and pointed the ORFS config to that file. This avoided repeating earlier SystemVerilog conversion issues while still allowing ORFS/Yosys to map the design to the ASAP7 standard-cell library.

Terminal - staging the final converted RV12 RTL

```
cd $HOME/OpenROAD-flow-scripts/flow
mkdir -p designs/asap7/rv12
mkdir -p designs/src/rv12
cp $HOME/OpenLane/designs/rv12/src/generated_rtl/rv12_converted.v    designs/src/rv12/rv12_converted.v

grep -n "module riscv_top_ahb3lite" designs/src/rv12/rv12_converted.v | head
```

The final synthesis top was riscv_top_ahb3lite. The clock port used in all SDC files was HCLK. This matches the AHB3-Lite integrated RV12 top-level wrapper used for the final ASAP7 frequency sweep.

4. Clock Constraints and Design Configuration

4.1 Per-frequency SDC files

The assignment formula is $\text{period_ns} = 1000 / \text{frequency_MHz}$. Because ASAP7 timing reports in this ORFS environment use ps as the default time unit, I wrote the SDC clock periods as 2000, 1667, and 1429.

Terminal - final RV12 SDC constraints

```
cat > designs/asap7/rv12/constraint_500.sdc << 'EOF'
create_clock -name HCLK -period 2000 [get_ports HCLK]
EOF

cat > designs/asap7/rv12/constraint_600.sdc << 'EOF'
create_clock -name HCLK -period 1667 [get_ports HCLK]
EOF

cat > designs/asap7/rv12/constraint_700.sdc << 'EOF'
create_clock -name HCLK -period 1429 [get_ports HCLK]
EOF
```

4.2 Final working config.mk files

One config.mk file was generated for each frequency. The important settings were the ASAP7 platform, top module name, the converted RV12 Verilog file, the per-frequency SDC, the fanout constraint, the increased memory synthesis limit, and a compact die/core area to avoid unnecessary tapcell explosion.

Terminal - representative config_500.mk

```
export PLATFORM = asap7
export DESIGN_NICKNAME = rv12_500
export DESIGN_NAME = riscv_top_ahb3lite
export VERILOG_FILES = ./designs/src/rv12/rv12_converted.v
export SDC_FILE = ./designs/asap7/rv12/constraint_500.sdc
export MAX_FANOUT = 8
export DIE_AREA = 0 0 200 200
export CORE_AREA = 2 2 198 198
export PLACE_DENSITY = 0.60
export TNS_END_PERCENT = 100
export SYNTH_MEMORY_MAX_BITS = 1000000
```

5. Issues Faced and How Each Was Fixed

#	Issue / problem	Root cause	Fix that worked
1	Older OpenLane/Sky130 result did not match final Week 4 target	Earlier run used 200/300/400 MHz and sky130A	Re-ran RV12 in ORFS with ASAP7 at 500/600/700 MHz
2	Synthesized memory size exceeded SYNTH_MEMORY_MAX_BITS	RV12 inferred memories exceeded ORFS default memory threshold	Added export SYNTH_MEMORY_MAX_BITS = 1000000
3	Large floorplan filled disk space	Initial 3000 x 3000 um die area caused 637,065 tapcells and huge ODB files	Deleted huge floorplan ODBs and reduced DIE_AREA to 200 x 200 um
4	No timing values directly after make synth	ORFS synth target creates mapped netlist/ODB but final STA report comes from floorplan	Ran make floorplan and used 2_floorplan_final.rpt
5	Many input/output delay warnings	Only a clock constraint was provided; I/O timing was not modeled in this assignment scope	Reported WNS/TNS from clocked internal paths and documented warning context

6. Running Synthesis at All Three Frequencies

After fixing the inferred-memory limit, synthesis completed successfully at 500 MHz, 600 MHz, and 700 MHz. The 500 MHz run showed Yosys mapping through ABC, then OpenROAD linking the final design and writing the synthesized database and SDC.

Terminal - synthesis command and success evidence

```
make DESIGN_CONFIG=./designs/asap7/rv12/config_500.mk clean_synth
make DESIGN_CONFIG=./designs/asap7/rv12/config_500.mk synth 2>&1 | tee /tmp/rv12_synth_500.log

# Success evidence from the completed run:
Design area 9964 um^2 100% utilization.
write_db ./results/asap7/rv12_500/base/1_synth.odb
write_sdc -no_timestamp ./results/asap7/rv12_500/base/1_synth.sdc
```

The same command pattern was then used for 600 MHz and 700 MHz. All three runs produced the expected results/asap7/rv12_<frequency>/base output tree, as well as synth_stat.txt and synth_check.txt under reports/asap7/rv12_<frequency>/base.

7. Metrics Extraction and Sources

The structural metrics came from synth_stat.txt. The timing metrics came from 2_floorplan_final.rpt, generated after make floorplan. WNS and TNS both reported 0.00 at all three frequencies, so the design met setup timing for each target frequency. The positive worst slack was taken from report_worst_slack inside the floorplan report.

Terminal - final metric extraction commands

```
for FREQ in 500 600 700; do
  STAT="reports/asap7/rv12_${FREQ}/base/synth_stat.txt"
  awk '/ cells$/ {print "Cell count:", $1} /Chip area/ {print} /sequential elements/ {print}' "$STAT"
  awk '/DFF.*ASAP7/ {sum += $1; print} END {print "Total FF count:", sum}' "$STAT"
  grep "ASAP7" "$STAT" | sort -k1,1nr | head -10
  grep -R "tns max\|wns max\|worst slack max" reports/asap7/rv12_${FREQ}/base/*floorplan*.rpt
done
```

8. Day 1 Metrics Table

Frequency	Clock Period	Cell Count	Chip Area	FF Count	Most Common Cell	TNS	WNS	Positive Setup Slack
500 MHz	2.000 ns	79,853	9964.496880 um ²	11,804	DFFHQNx1_ASAP7_75t_R = 10,388	0.00	0.00	+0.649 ns
600 MHz	1.667 ns	79,853	9964.496880 um ²	11,804	DFFHQNx1_ASAP7_75t_R = 10,388	0.00	0.00	+0.316 ns
700 MHz	1.429 ns	79,853	9964.496880 um ²	11,804	DFFHQNx1_ASAP7_75t_R = 10,388	0.00	0.00	+0.078 ns

The synthesized cell count, area, flip-flop count, and most common cell remained unchanged across 500, 600, and 700 MHz. This happened because the same final mapped netlist structure was sufficient for all three frequency targets, and timing was still met at 700 MHz.

Cell type	Count	Area contribution
DFFHQNx1_ASAP7_75t_R	10,388	~3.03E+03
AND3x1_ASAP7_75t_R	9,189	803.854
AOI21x1_ASAP7_75t_R	9,169	~1.07E+03
BUFx2_ASAP7_75t_R	8,011	584.002
NAND2x1_ASAP7_75t_R	7,057	617.346
AND2x2_ASAP7_75t_R	6,377	557.860
AO21x1_ASAP7_75t_R	5,297	463.382
INVx1_ASAP7_75t_R	5,012	219.225

9. Getting STA Numbers - make floorplan

The synth target alone did not provide the final WNS/TNS table used in this report. I therefore ran make floorplan for all three frequencies. Floorplan generated 2_floorplan_final.rpt, which provided tns max, wns max, worst slack max, and the detailed startpoint/endpoint path.

Terminal - floorplan command and final success evidence

```
for FREQ in 500 600 700; do
  echo "Running floorplan for rv12_$FREQ"
  make DESIGN_CONFIG=./designs/asap7/rv12/config_${FREQ}.mk floorplan 2>&1 | tee /tmp/rv12_floorplan_${FREQ}.log
done

# 700 MHz final success:
Inserted 1450 endcaps.
Inserted 2544 tapcells.
Design area 10145 um^2 26% utilization.
cp ./results/asap7/rv12_700/base/2_4_floorplan_pdn.odb ./results/asap7/rv12_700/base/2_floorplan.odb
```

10. Critical Path Analysis and Maximum Frequency

The repeated setup-critical path for all three frequencies was inside the RV12 multiplier unit. The startpoint was a multiplier bubble register and the endpoint was bit 30 of the multiplier result register. This indicates that the execution-stage multiplier datapath is the main timing-sensitive region of this synthesized RV12 implementation.

Frequency	Startpoint	Endpoint	Arrival / Required	Slack
500 MHz	mul_bubble_o\$_DFF_PN1_	mul_r_o[30]\$_DFF_P_	1336.25 / 1984.83 ps	+648.57 ps
600 MHz	mul_bubble_o\$_DFF_PN1_	mul_r_o[30]\$_DFF_P_	1336.25 / 1651.83 ps	+315.57 ps
700 MHz	mul_bubble_o\$_DFF_PN1_	mul_r_o[30]\$_DFF_P_	1336.25 / 1413.83 ps	+77.57 ps

At 700 MHz the positive slack is only +77.57 ps. Since the data arrival time is 1336.25 ps and the required time is 1413.83 ps, RV12 still passes the target, but the design is close to its timing limit. A rough estimate from this timing path suggests the maximum frequency is slightly above 700 MHz, around the low-to-mid 700 MHz range, before setup timing would likely start failing.

11. RV12 Design-Specific Challenges and Questions

11.1 Which part of RV12 dominates the critical path?

The timing report shows the critical setup path in the multiplier unit, from core.ex_units.genblk1.mul.mul_bubble_o to core.ex_units.genblk1.mul.mul_r_o[30]. This means the multiply datapath is the most important frequency-scaling path in this synthesis result.

11.2 What does the flip-flop count say about the design?

RV12 used 11,804 flip-flops, dominated by 10,388 DFFHQNx1_ASAP7_75t_R cells and 1,416 DFFASRHQNx1_ASAP7_75t_R cells. This high sequential count is expected for a pipelined CPU core with register files, control/status storage, queues, memory interface logic, and pipeline state registers.

11.3 Why did inferred memory create a synthesis issue?

The RV12 design includes inferred memories/register arrays. ORFS initially rejected the synthesis because the inferred memory size exceeded the default SYNTH_MEMORY_MAX_BITS limit. Increasing the limit to 1000000 allowed the memories to be synthesized and mapped for this experiment.

11.4 What should be improved if the design fails above 700 MHz?

The first improvement target should be the multiplier path. Possible fixes include adding or improving multiplier pipelining, changing multiplier implementation style, tightening synthesis strategy, or applying physical optimization after placement and routing. Since the design still passes 700 MHz with only +77.57 ps slack, small extra delays from routing could become important in later flow stages.

12. Frequency Scaling Observations

Frequency	Clock period	Positive slack	Observation
500 MHz	2.000 ns	+0.649 ns	Large timing margin
600 MHz	1.667 ns	+0.316 ns	Still comfortable, but reduced margin
700 MHz	1.429 ns	+0.078 ns	Meets timing but close to limit

The trend is clear: cell count and area are stable, but timing slack decreases as the clock period becomes smaller. RV12 still meets timing at 700 MHz, but the 700 MHz slack is much smaller than the 500 MHz and 600 MHz cases. This suggests that the design is practical at the assigned targets, but later placement/routing or extra constraints may require additional optimization.

13. Deliverables and GitHub Submission Packaging

The final GitHub repository contains both RV12 and NPU outputs. The RV12 results were pushed under rv12/ with configuration files, constraints, synthesis reports, floorplan reports, logs, generated netlists, and a short analysis markdown file. This makes the submission reproducible because the exact configs, SDCs, logs, and source-to-netlist outputs are available.

Repository item	Purpose
rv12/configs/	Per-frequency ORFS config.mk files
rv12/constraints/	Per-frequency create_clock SDC files
rv12/reports/synthesis/	synth_stat.txt and synth_check.txt for metrics
rv12/reports/floorplan/	2_floorplan_final.rpt for WNS/TNS and critical path
rv12/logs/	Yosys and OpenROAD logs for evidence
rv12/results/netlist/	Mapped Verilog netlists and generated SDC files
rv12/RV12_frequency_analysis.md	Short summary of final metrics and issues

14. Key Facts Learned This Week

- ASAP7 timing values in this ORFS setup are reported in ps, so the SDC period values were written as 2000, 1667, and 1429.
- The same mapped RV12 netlist met 500, 600, and 700 MHz, but the margin reduced strongly with increasing frequency.
- For this RV12 run, the multiplier datapath was the setup-critical path, not the AHB wrapper or memory queue path.
- Large die/core settings can generate excessive tapcells and fill disk space, even when the real design cell area is much smaller.
- WNS/TNS should be taken from the floorplan STA report in this ORFS flow, not only from the synth target.

15. Reflection and Recommendation / Future Scope

The RV12 frequency sweep was more challenging than the NPU run because the design is much larger and contains inferred memories, a large number of sequential cells, and a timing-sensitive multiplier datapath. The biggest practical lesson was that a successful synthesis result is not enough; the report needs traceable evidence from logs, synth_stat.txt, and floorplan timing reports.

For future improvement, RV12 should be taken further through placement, CTS, and routing to see whether the +77.57 ps margin at 700 MHz remains positive after real routing parasitics. If timing degrades, the multiplier datapath should be the first optimization target. Possible approaches include deeper multiplier pipelining, arithmetic restructuring, or using a faster implementation strategy for the execution-stage multiplier.